

SECTION 6 – INTERCONNECT DATA

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
6-1	INTRODUCTION	6-3
6-1-1	Component Designators	6-4
6-1-2	Definition of Terms	6-5
6-2	SYSTEM INTERCONNECTS ADDED BY UPGRADES	6-6
6-2-1	Introduction	6-6
6-2-2	M1033K – Indigo to Octane Computer Upgrade	6-8
6-2-3	M1033L – Indigo to Octane II with EXCITE Upgrade	6-8
6-2-4	M1060JW – Shield/Cryo Cooler Compressor	6-9
6-2-5	M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array	6-10
6-2-6	M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector	6-17
6-2-7	M3044TA – Site Collector TwinSpeed Fixed Site With Phased Array Long Cables	6-23
6-2-8	M3044TC – EXCITE TwinSpeed Fixed Site Cable Collector Long Cables	6-23
6-2-9	M3060TA – 1.5T Magnet With TwinSpeed Enclosure	6-24
6-2-10	M3063TA – Site Collector TwinSpeed Transportable/Relocatable With Phased Array	6-25
6-2-11	M3063TC – EXCITE TwinSpeed T/R Cable Collector	6-25
6-2-12	M3080TA – 2nd Order Resistive Shim Option	6-25
6-2-13	M3088TA – Outdoor TSCC System Interconnects	6-26
6-2-14	M3088TB – Indoor Water Cooled TSCC System Interconnects	6-31
6-2-15	M3088TC – Indoor Air Cooled TSCC System Interconnects	6-33
6-2-16	M3088TE – Indoor Water Cooled TGWC System Interconnects	6-35
6-2-17	M3088TF – Outdoor TGWC System Interconnects	6-37
6-2-18	M3088TF – Outdoor TGWC System Interconnects	6-40
6-2-19	M3088TH – GE Main Disconnect Panel	6-42
6-2-20	M3090AH – Horizon to TwinSpeed Upgrade System Electronics	6-42
6-2-21	M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet	6-43
6-2-22	M3090EB – EXCITE UPGRADE KIT	6-44
6-2-23	M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-45
6-2-24	M3090TA – Twin Upgrade For LCC Horizon Fixed & T/R	6-46
6-2-25	M3090TE – Covers From WideOpen For Fixed & T/R Sites	6-47

TABLE OF CONTENTS (Continued)

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
6-3	ADVANTAGE WORKSTATION INTERCONNECTS	6-48
6-4	INTERCONNECTS RE-ROUTED BY ACGD UPGRADES	6-48
6-5	INTERCONNECTS REMOVED BY SIGNA 1.5T TWINSPEED WITH EXCITE TECHNOLOGY UPGRADES	6-48
6-5-1	Signa MR/i, CV/i, NV/i & Signa LX, CVMR System With LCC Magnet & ACGD & SRF Cabinets	6-48
6-5-2	Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinets	6-52
6-5-3	Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets	6-56
6-5-4	Signa LX & CVMR Without LCC Magnet Installed	6-60
6-5-5	Signa Horizon 5.X System	6-60
6-5-6	Signa Advantage 5.X & 4.X System	6-60
6-6	INTERCONNECTS REMOVED BY SIGNA 1.5T TWINSPEED UPGRADES	6-61
6-6-1	Signa MR/i, CV/i, NV/i & Signa LX, CVMR System With LCC Magnet & ACGD & SRF Cabinets	6-61
6-6-2	Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinets	6-63
6-6-3	Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets	6-66
6-6-4	Signa LX & CVMR Without LCC Magnet Installed	6-69
6-6-5	Signa Horizon 5.X System	6-69
6-6-6	Signa Advantage 5.X & 4.X System	6-69
6-7	REAR PEDESTAL AND ENCLOSURE CABLE ACCESS	6-70
6-8	SPECTROSCOPY OPTION INTERCONNECTS	6-71
6-8-1	M1033MD – MNS Kit For Signa LX Systems	6-71
6-8-2	M3090DB – 5.5 MNS Upgrade Kit For ACGD Site	6-73
6-8-3	M3090DC – LX MNS Upgrade Kit For ACGD Site	6-73

6-1 INTRODUCTION

Section 6, INTERCONNECT DATA, addresses system interconnections which are impacted by the Signa 1.5T TwinSpeed upgrade catalogs. The subsections are broken down as follows:

- 6-1 INTRODUCTION
 - general upgrade information, component designations
- 6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES
 - cable interconnects for the system provided by Horizon LX with ACGD Upgrade catalogs
- 6-3 ADVANTAGE WORKSTATION INTERCONNECTS
 - Refer to Direction 2261309-100 *Advantage Workstation Ultra60 Pre-Installation* for AW interconnect information.
- 6-4 INTERCONNECTS RE-ROUTED BY ACGD UPGRADES
 - existing system cable interconnects re-labeled and re-routed during ACGD Upgrades if not present prior to Signa 1.5T TwinSpeed upgrade.
- 6-5 INTERCONNECTS REMOVED BY SIGNA 1.5T TWIN SPEED WITH EXCITE TECHNOLOGY UPGRADES
 - existing system cable interconnects removed during Signa 1.5T TwinSpeed With EXCITE Technology upgrade.
- 6-6 INTERCONNECTS REMOVED BY SIGNA 1.5T TWIN SPEED UPGRADES
 - existing system cable interconnects removed during Signa 1.5T TwinSpeed upgrade.
- 6-7 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
 - need to plan Rear Pedestal cable access and routing due to increase of cabling and hoses
- 6-8 SPECTROSCOPY OPTION INTERCONNECTS
 - additional cable interconnects for the system provided by Spectroscopy Option Upgrade catalogs for Signa 1.5T TwinSpeed Upgrades.

6-1-1 Component Designators

The Component Designator System is used as the means of identifying system components in a consistent manner. All subsystem cabinets and other components are referred to by their component designators in the diagrams and tables of this section. (For example, the System Cabinet is referred to as MR2.) Refer to Table 6-1 for component designators impacted by Signa 1.5T TwinSpeed or TwinSpeed With EXCITE Technology upgrade.

TABLE 6-1
COMPONENT DESIGNATION

APPLICABLE SYSTEM	COMPONENT DESIGNATOR	DESCRIPTION
Signa 1.5T TwinSpeed System	EO1/EO2	Emergency Off Buttons
	MDP	Main Disconnect Panel
	MG2	Magnet Enclosure
	MG3	Magnet Rear Pedestal
	MG6	Blower Box
	MR1	SRF Cabinet
	MR2	System Cabinet for TwinSpeed or TwinSpeed With EXCITE Technology
	MR3	ACGD/PDU Cabinet has PD1 in lower portion cabinet
	MS1	Superconducting Magnet
	MS4	Magnet Rundown Unit
	MS5	Shield/Cryo Cooler Compressor Cabinet
	MSM1	Magnet Monitor
	MSM3	Modem for Magnet Monitor
	OW1	Operator Workspace
	PD1	Power Distribution Unit (PDU) is a module in lower portion of ACGD/PDU Cabinet
	PP1	Penetration Panel
	PT1	Patient Transport Table
	RCP	Remote Control Box for Outdoor TSCC, Outdoor TGWC, & Water Cooled TGWC
	TAC	Twin Accessory Cabinet
	TSCC	Twin System Cooling Cabinet (provides water cooling for Shield/Cryo Cooler Compressor & Gradient Coil)
	TGWC	Twin Gradient Water Cooler (provides water cooling for Gradient Coil ONLY; Customer supplied water cooling is required for Shield/Cryo Cooler Compressor Cabinet)
System Options	MNS	MNS Cabinet
	AW	Advantage Workstation
	MSM4	UPS for Magnet Monitor
	MUX	Phone Line Multiplexer for Magnet Monitor
Signa LX with ACGD or Signa Horizon LX Obsolete Equipment	HE1	Heat Exchanger
	MR1	RF/PDU Cabinet RF/Penetration Cabinet RF Cabinet
	MR3	SGD Base Gradient Cabinet for Base System or SGD Hi-Slew Gradient Cabinet for HiSpeed or EchoSpeed 8645 Gradient Cabinet Gradient Amp Cabinet
	PD1	Power Distribution Unit (PDU) is a module in lower portion of RF/PDU Cabinet Full Size (150 KVA or 75 KVA) or Compact Power Distribution Unit
	PP2	Penetration Cabinet
	WC1	Gradient Coil Water Chiller

6-1-2 Definition of Terms

The definition of terms used in following sections tables are:

- Group Number: identifying number referenced to bundles (i.e. groups) of cables as shown in interconnect diagrams
- Between Units (From/To): component designators as found in tables
- Usable Length: total length of a cable MINUS any required take up within cabinets
- Run Number: unique number assigned to each GE-supplied cable

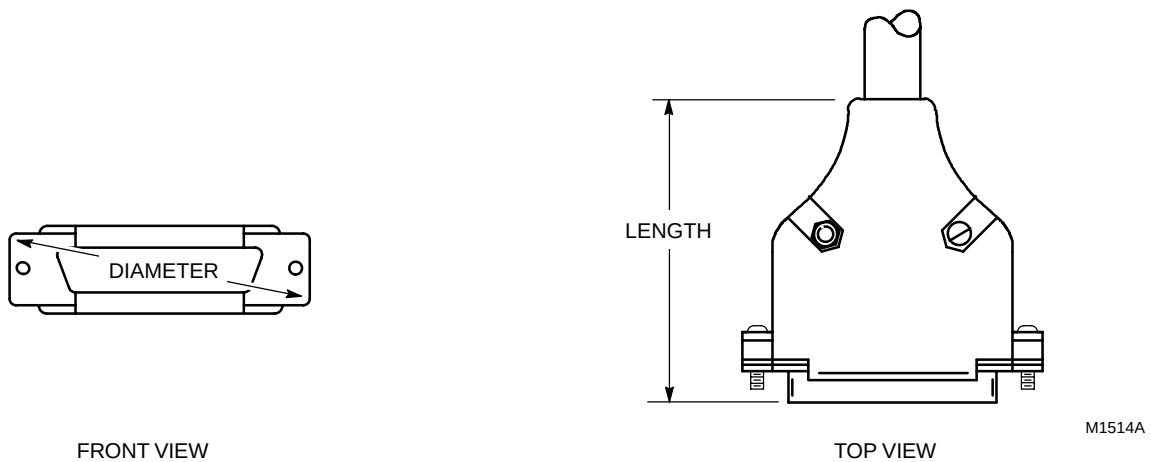
Note

The run number must be used when making special cable orders.

- Cable Diameter: diameter of an individual cable
- Plug Pulling Diameter x Length: cable plug dimensions

Note

In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 -x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). Of course, the same number of connectors apply to the metric dimensions as well.



SUBMINIATURE-D CONNECTOR PLUG PULLING DIMENSIONS
ILLUSTRATION 6-1

6-2 SYSTEM INTERCONNECTS ADDED BY ACGD UPGRADES

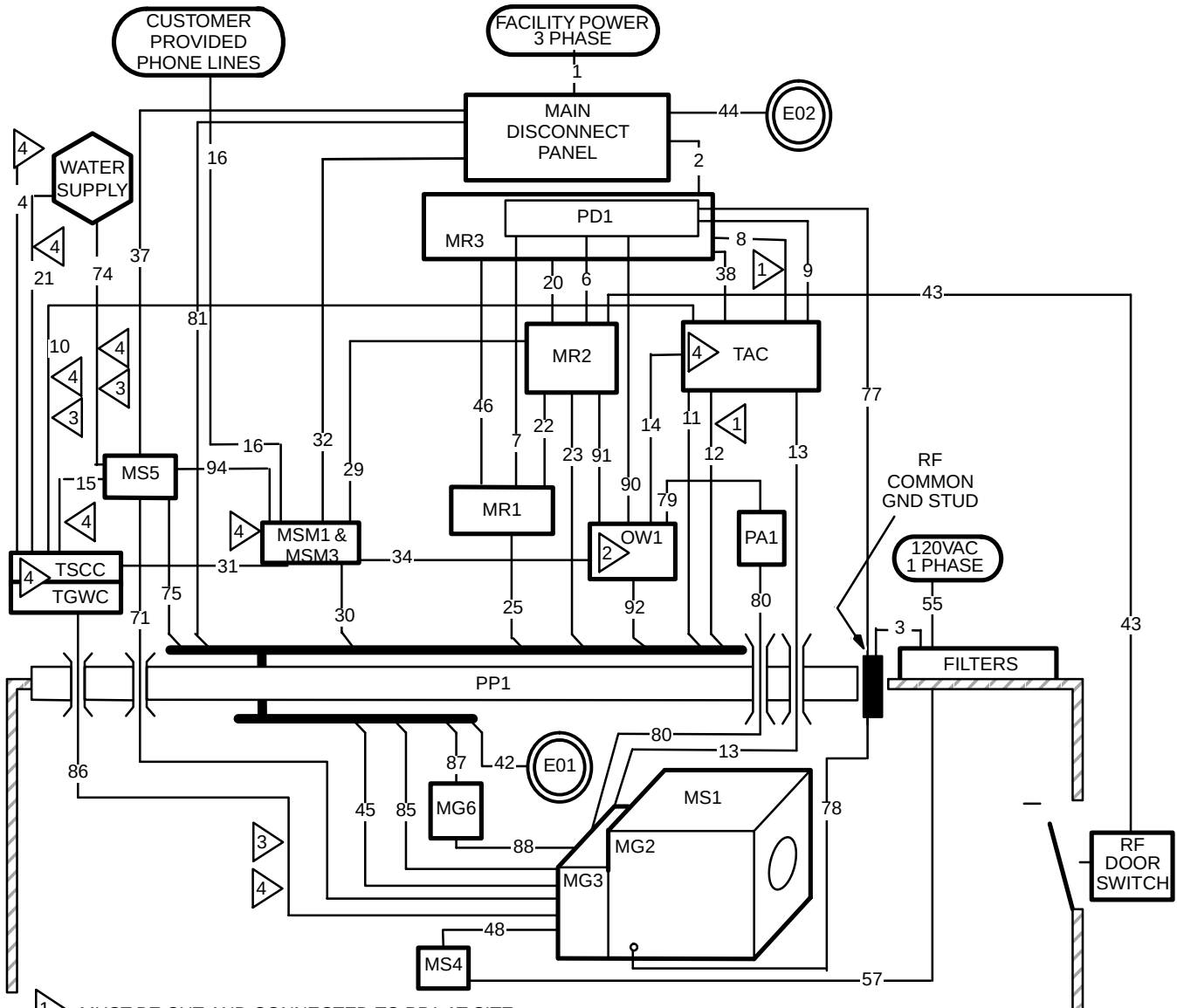
6-2-1 Introduction

Illustration 6-2 interconnect diagrams shows the groups to which cables are added by the various Signa 1.5T TwinSpeed or TwinSpeed With EXCITE Technology upgrade catalogs. The following sub-sections list the cables provided by the various upgrade catalogs. Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Unless otherwise specified, cables and components listed in tables in this section are supplied by GE.

Note

Need to verify customer supplied Group1 (Facility Power to Main Disconnect Control) and Group 2 (Main Disconnect Control) are properly sized for specifications in Section 5, Power Requirements and in accordance with local and national codes.

6-2-1 Introduction (Continued)



- 1 MUST BE CUT AND CONNECTED TO PP1 AT SITE.
NOTE: IMPEDANCE IS NOT CRITICAL SO EXCESS CABLE SHOULD BE CUT OFF.
- 2 OPERATOR WORKSPACE (OW1) SUBSYSTEM EQUIPMENT IS PROVIDED WITH MAXIMUM LENGTH CABLES POSSIBLE. SEVERAL OW ASSEMBLIES ARE MOUNTED TO OW TABLE & OW INTERCONNECTS ARE ROUTED THROUGH TABLE CABLE TRAY. FOR REFERENCE USE ONLY THE OW1 RUN NUMBERS ARE 049, 792, 793, 794, 795, 796, 797, 798, 799, 806, & 807.
- 3 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- 4 FOR GROUPS 4, 10, 15, 21, 31, 49, 50, 51, 53, 74, & 86 REFER TO APPROPRIATE SECTION FOR TSCC OR TGWC INTERCONNECTS (SECTION 6-2-13 through 6-2-18) FOR ADDITIONAL AND CONFIGURATION APPROPRIATE INTERCONNECTS.

SIGNA 1.5T TWINSPEED SYSTEM UPGRADE INTERCONNECT DIAGRAM

ILLUSTRATION 6-2

6-2-2 M1033K – Indigo to Octane Computer Upgrade

M1033K provides interconnects for an Indigo to Octane Computer Upgrade listed in Table 6-2.

TABLE 6-2
CABLES PROVIDED BY M1033K

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 1	818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).
Note 1 M3044TA can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).								

6-2-3 M1033L – Indigo to Octane II with EXCITE Upgrade

M1033L provides interconnects for an Indigo to Octane II with EXCITE Upgrade listed in Table 6-3.

TABLE 6-3
CABLES PROVIDED BY M1033L

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 1	1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
			1032	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)		
Note 1 M3044TC can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).								

6-2-4 M1060JW – Shield/Cryo Cooler Compressor

M1060JW provides interconnects for Shield/Cryo Cooler Compressor for the LCC Magnet listed in Table 6-4.

TABLE 6-4
CABLES PROVIDED BY M1060JW

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
37	MDP	MS5	29.8 (9.1) minus takeup at MDP	—	1.29 (32.75)	Hard Wired	Hard Wired	Both ends have liquid tight connector which wires pass through for hard wire connections.
45	PP1	MG3	30 (9.15)	624	0.26 (6.6)	Ring Terminals	1.00x1.38 (25.4x35.1)	
71	MS5	MS1	38.5 (11.7)	621 622	1.65 (41.9) 1.65 (41.9)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	Runs 621 and 622 are continuous helium lines routed through PP1. Bend radius of Runs 621 and 622 is 8 in. (203.2 mm). Cable diameter includes foam insulation installed on lines.
75	MS5	PP1	42 (12.8)	623	0.21 (5.28)	1.00x1.38 (25.4x35.1)	Ring Terminals	

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array

M3043TA provides standard length system interconnects for fixed site listed in Table 6-5. There are also requirements for customer supplied interconnects as indicated in Table 6-7 .

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
3	Facility Emerg Power Filter	PP1	—	—	—	—	—	Customer Furnished Ground.
6	PD1 See Note 1	MR2	22 (6.70)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
				972	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
7	PD1 See Note 1	MR1	22 (6.70)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				967	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
9	PD1 See Note 1	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	
10	RCP, TSCC, or TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable Refer to appropriate section for TSCC or TGWC interconnects (Section 6–2–13 through 6–2–18) for details & to determine FROM connection.
11	TAC	PP1	21 (6.40)	991	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
12	TAC	PP1	33 (10.0)	979	1.21 (30.7)	Hard Wired	Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
			33 (10.0)	980	1.21 (30.7)	Hard Wired	Ring Terminal	
				981	1.21 (30.7)	Hard Wired	Ring Terminal	
				988	0.26 (6.6)	Ring Terminal	Ring Terminal	Ground wire
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	14 (4.27)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
23	MR2	PP1	31 (9.45)	231	0.212 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
			See Note 2	233/ 499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
			488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)		
			489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)		
			490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)		
			711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)		
Note 2 M3044TA can be ordered to increase usable length of Group 23 to 56 ft (17.1 m).								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40) See Note 3	044	0.20 (5.1)	Ring Terminal	Ring Terminal	Ground cable Has a minimum bend radius of 10 in. (254 mm). Has a minimum bend radius of 5 in. (127 mm).
			235	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)		
			236	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)		
			486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
			487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)		
			726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)		
			768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
			771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			773	0.34 (8.64)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals		
38	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
Note 3 M3044TA can be ordered to increase usable length of Group 25 to 51 ft (15.5 m) except for Runs 235 & 236 which may need Special Order Inquiry (SOI) to provide longer length.								
(Continued)								

ARCHITECTURAL/INSTALLATION PLANNING USE ONLY

GE MEDICAL SYSTEMS

SIGNA 1.5T TWINSPEED UPGRADE PRE-INSTALLATION

REV 1

DIRECTION 2304880

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
43	MR2	RF Door Switch	67 (20.4) minus takeup at RF Door Switch See Note 4	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor.
44	MDP	EO2	—	—	—	—	—	Customer Furnished.
45	PP1	MG3	30 (9.15)	257	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Has a minimum bend radius of 10 in. (254 mm).
				258	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Has a minimum bend radius of 5 in. (127 mm).
				262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
Note 4 M3044TA can be ordered to increase usable length of Group 43 to 97 ft (29.6 m) minus takeup at RF Door Switch.								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45 (contin- ue)	PP1	MG3	30 (9.15)	727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				990	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
46	MR1	MR3	14 (4.27) See Note 5	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is flexible conduit containing fiber optic cables.
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud See Note 6	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
79	OW1	PA1	5 (1.5) minus takeup at PA1	—	0.13 (3.2)	3.00x3.00 (76.2x76.2)	0.38x1.75 (9.6x44.5)	
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
81	MDP	PP1	40 (12.2)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	
Note 5 M3044TA can be ordered to increase usable length of Group 46 to 59 ft (18.0 m).								
6 M3044TA can be ordered to increase usable length of Group 78 to 91 ft (27.7 m) minus takeup at RF Common GND Stud.								
(Continued)								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
85	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
				983	1.21 (30.7)	Ring Terminals	Hard Wired	
				984	1.21 (30.7)	Ring Terminals	Hard Wired	
				985	1.21 (30.7)	Ring Terminals	Hard Wired	
				986	1.21 (30.7)	Ring Terminals	Hard Wired	
				987	1.21 (30.7)	Ring Terminals	Hard Wired	
86 See Note 7	TSCC/ TGWC See Notes	MG2	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6–2–13 through 6–2–18) for details & to determine FROM connection.
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes & Note 8	784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 being mounted on floor and floor routing of cables.
88	MG6	MG3	13 (3.95) minus takeup	—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.
90	PD1 See Note 1	OW1	51 (15.5) See Note 9	048	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				969	0.70 (17.78)	Hard Wired	Hard Wired	#10 AWG / 4 wire power cable
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1). 7 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 8 M3044TA can be ordered to increase usable length of Group 87 to 71 ft (21.6 m). 9 M3044TA can be ordered to increase usable length of Group 90 to 84 ft (25.6 m).								

6-2-5 M3043TA – Site Collector TwinSpeed Fixed Site With Phased Array (Continued)

TABLE 6-5
CABLES PROVIDED BY M3043TA & M3044TA (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 10	789	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8). Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).
			791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes		
			836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)		
92	PP1	OW1	53 (16.1) See Note 11	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note 10 M3044TA can be ordered to increase usable length of Group 91 to 81 ft (24.7 m). 11 M3044TA can be ordered to increase usable length of Group 92 to 88 ft (26.8 m).								

6-2-6 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector

M3043TC provides standard length system interconnects for fixed site listed in Table 6-6. There are also requirements for customer supplied interconnects as indicated in Table 6-6.

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
3	Facility Emerg Power Filter	PP1	—	—	—	—	—	Customer Furnished Ground.
6	PD1 See Note 1	MR2	22 (6.70)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
7	PD1 See Note 1	MR1	22 (6.70)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				967	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
9	PD1 See Note 1	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	
10	RCP, TSCC, or TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
11	TAC	PP1	21 (6.40)	991	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
12	TAC	PP1	33 (10.0)	979	1.21 (30.7)	Hard Wired	Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
			33 (10.0)	980	1.21 (30.7)	Hard Wired	Ring Terminal	
				981	1.21 (30.7)	Hard Wired	Ring Terminal	
				988 X 3	0.26 X 3 (6.6 X 3)	Ring Terminal	Ring Terminal	A Ground wire for each of Runs 979, 980, & 981
13	TAC	MG2/3	80 (24.4)	—	1.27 (32.3)	Flexible Line	Flexible Line	Run is routed through waveguide in PP1, cut to length at site.
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	1034	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	14 (4.27)	229	0.23 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Run 1033 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
				1033	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	
23	MR2	PP1	31 (9.45) See Note 2	231	0.212 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
				233/ 499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1025	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1026	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1027	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1028	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1029	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
				1030	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
				1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	

Note 2 M3044TC can be ordered to increase usable length of Group 23 to 56 ft (17.1 m).

(Continued)

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40) See Note 3	044	0.20 (5.1)	Ring Terminal	Ring Terminal	Ground cable Has a minimum bend radius of 10 in. (254 mm). Has a minimum bend radius of 5 in. (127 mm).
			235	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)		
			236	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)		
			726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)		
			768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
			771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
			773	0.34 (8.64)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
			783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals		
38	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
Note 3 M3044TA can be ordered to increase usable length of Group 25 to 51 ft (15.5 m) except for Runs 235 & 236 which may need Special Order Inquiry (SOI) to provide longer length.								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
43	MR2	RF Door Switch	67 (20.4) minus takeup at RF Door Switch See Note 4	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor.
44	MDP	EO2	—	—	—	—	—	Customer Furnished.
45	PP1	MG2/3	30 (9.15)	257	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Has a minimum bend radius of 10 in. (254 mm).
				258	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Has a minimum bend radius of 5 in. (127 mm).
				262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
Note 4 M3044TA can be ordered to increase usable length of Group 43 to 97 ft (29.6 m) minus takeup at RF Door Switch.								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45 (contin- ue)	PP1	MG2/3	30 (9.15)	781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				990	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				1036	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1037	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1038	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1039	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1040	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1041	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1042	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud See Note 5	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
81	MDP	PP1	40 (12.2)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	
Note 5 M3044TA can be ordered to increase usable length of Group 78 to 91 ft (27.7 m) minus takeup at RF Common GND Stud.								
(Continued)								

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
85	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
				983	1.21 (30.7)	Ring Terminals	Hard Wired	
				984	1.21 (30.7)	Ring Terminals	Hard Wired	
				985	1.21 (30.7)	Ring Terminals	Hard Wired	
				986	1.21 (30.7)	Ring Terminals	Hard Wired	
				987	1.21 (30.7)	Ring Terminals	Hard Wired	
86 See Note 6	TSCC/ TGWC See Notes	MG2	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes & Note 7	045	0.195 (4.95)	Ring Terminal	Ring Terminal	The usable length is based on MG6 being mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 being mounted on floor and floor routing of cables.
88	MG6	MG3	13 (3.95) minus takeup	—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.
90	PD1 See Note 1	OW1	51 (15.5) See Note 8	048	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				969	0.70 (17.78)	Hard Wired	Hard Wired	#10 AWG / 4 wire power cable

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

6 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

7 M3044TC can be ordered to increase usable length of Group 87 to 71 ft (21.6 m).

8 M3044TC can be ordered to increase usable length of Group 90 to 84 ft (25.6 m).

6-2-5 M3043TC – EXCITE TwinSpeed Fixed Site Cable Collector (Continued)

TABLE 6-6
CABLES PROVIDED BY M3043TC & M3044TC (Continued)

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5) See Note 9	789	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
			1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)		
			1032	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		
			1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)		
92	PP1	OW1	53 (16.1) See Note 10	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note 9 M3044TC can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).								
Note 10 M3044TC can be ordered to increase usable length of Group 92 to 88 ft (26.8 m).								

6-2-7 M3044TA – Site Collector TwinSpeed Fixed Site With Phased Array Long Cables

M3044TA provides longer length system interconnects for fixed site as indicated in Table 6-5 notes.

6-2-8 M3044TC – EXCITE TwinSpeed Fixed Site Cable Collector Long Cables

M3044TC provides longer length system interconnects for fixed site as indicated in Table 6-6 notes.

6-2-9 M3060TA – 1.5T Magnet With TwinSpeed Enclosure

M3060TA provides 1.5T Magnet system interconnects listed in Table 6-7. There are also requirements for customer supplied interconnect as indicated in Table 6-7 .

TABLE 6-7
CABLES PROVIDED BY M3080TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
16	Phone Line Connection	MSM1	—	—	—	—	—	Refer to Section 2-6-6 Telephone Lines Requirements for additional customer phone line information.
29	MSM1	MR2	54 (16.4) minus takeup at MSM1	823	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
30	MSM1	PP1	54 (16.4) minus takeup at MSM1	824	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				825	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				914	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
31	MSM1	TSCC/ TGWC	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
32	MDP	MSM1 & MSM3	6 (1.8) minus takeup at each end	—	0.375 (9.5)	1.2x3.00 (30.5x76.2)	1.25x2.5 (31.8x63.5)	MSM1 power cable
				—	0.25 (6.35)	2.00x2.50 (50.8x63.5)	2.00x2.50 (50.8x63.5)	Modem for MSM1 power cable. Modem located on top of MSM1. Refer to Section 2-6-5
34	MSM1	OW1	70 (21.3) minus takeup at MSM1	942	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
45	PP1	MG3	30 (9.15)	828	0.31 (7.75)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				829	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
48	MS4	MS1	86 (26.2)	606	0.30 (7.6)	0.65x1.85 (16.5x47.0)	0.65x1.85 (16.5x47.0)	
55	Facility Power	Filter	—	—	—	—	—	Customer Furnished Magnet Room power (refer to Sections 5-1 and 7-4).
57	Filter	MS4	—	—	—	—	—	Customer Furnished (refer to Sections 5-1).

6-2-9 M3060TA – 1.5T Magnet With TwinSpeed Enclosure (Continued)TABLE 6-7 (Continued)
CABLES PROVIDED BY M3080TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
77	PD1 See Note 1	RF COM- MON GND STUD	—	—	0.584 (14.8)	Ring Terminal	Ring Terminal	Customer supplied ground cable, refer to Section 5-4-2.
94	MSM1	MS5	54 (16.4) minus takeup at MSM	826	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

6-2-10 M3063TA – Site Collector TwinSpeed Transportable/Relocatable With Phased Array

M3063TA provides Transportable/Relocatable appropriate lengths of system interconnects similar to M3043TA. Upgrade plans are available from the van manufacturer.

6-2-11 M3063TC – EXCITE TwinSpeed T/R Cable Collector

M3063TC provides Transportable/Relocatable appropriate lengths of system interconnects similar to M3043TC. Upgrade plans are available from the van manufacturer.

6-2-12 M3080TA – 2nd Order Resistive Shim Option

M3080TA provides system interconnects for the 2nd Order Resistive Shim Option listed in Table 6-8.

TABLE 6-8
CABLES PROVIDED BY M3080TA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
11	TAC	PP1	21 (6.40)	611	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 611 for system with 2nd Order Resistive Shim Option
14	TAC	OW1	25 (7.62)	568	0.36 (9.14)	1.30x2.00 (33.5x50.8)	2.30x2.10 (58.5x53.5)	Used for 2nd Order Resistive Shim Option
45	PP1	MG3	30 (9.15)	612	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 612 for system with 2nd Order Resistive Shim Option

6-2-13 M3088TA – Outdoor TSCC System Interconnects**Note**

Refer to **Section 2-5-1 TSCC Configurations Siting Considerations** Table 2-4 for listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 470 Pre installation Installation Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for the Outdoor TSCC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Outdoor TSCC must be in accordance with local and national codes.**

Vertical Separation Requirements

- Outdoor TSCC must not exceed 100 ft (30.5 m) vertical separation from the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements

Customer must provide water supply and return lines between the Outdoor TSCC and the GE supplied hoses for the Shield/Cryo Cooler Compressor.

- Ellis And Watts provides two 0.75 in. (19 mm) FNPT fittings at the Outdoor TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2-10 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 0.75 in. (19 mm) MNPT connections and 0.75 in. (19 mm) ID copper lines.
- Ellis And Watts provides two 0.5 inch (12.7 mm) ID rubber hoses of 60 ft (18.3 m) total length [usable length of 55 ft (16.7 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Shield/Cryo Cooler Compressor Cabinet. Site layout for the Shield/Cryo Cooler Compressor water cooling lines and hoses must meet the following:
 - Total line length from the Outdoor TSCC connection to the Shield/Cryo Cooler Compressor connection **MUST NOT EXCEED** 160 ft (48.8 m). Ellis And Watts supplied rubber hoses for Shield/Cryo Cooler Compressor is a maximum length of 60 ft (18.3 m) total length [55 ft (16.8 m) usable length].
 - If the 0.75 in. (19 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
 - Copper lines must be insulated outdoors.

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)**Gradient Coil Water Cooling Lines & Hoses Requirements**

- Ellis And Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2-10 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
 - Total line length from the Outdoor TSCC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TSCC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
 - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).

RCP Data Cables Requirements

- Ellis & Watts provides three data cables that connect between the RCP and the Outdoor TSCC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TSCC and RCP placement [total length minus height of connection at Outdoor TSCC and height of connection at RCP, no cable takeup inside of TSCC or RCP].

Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

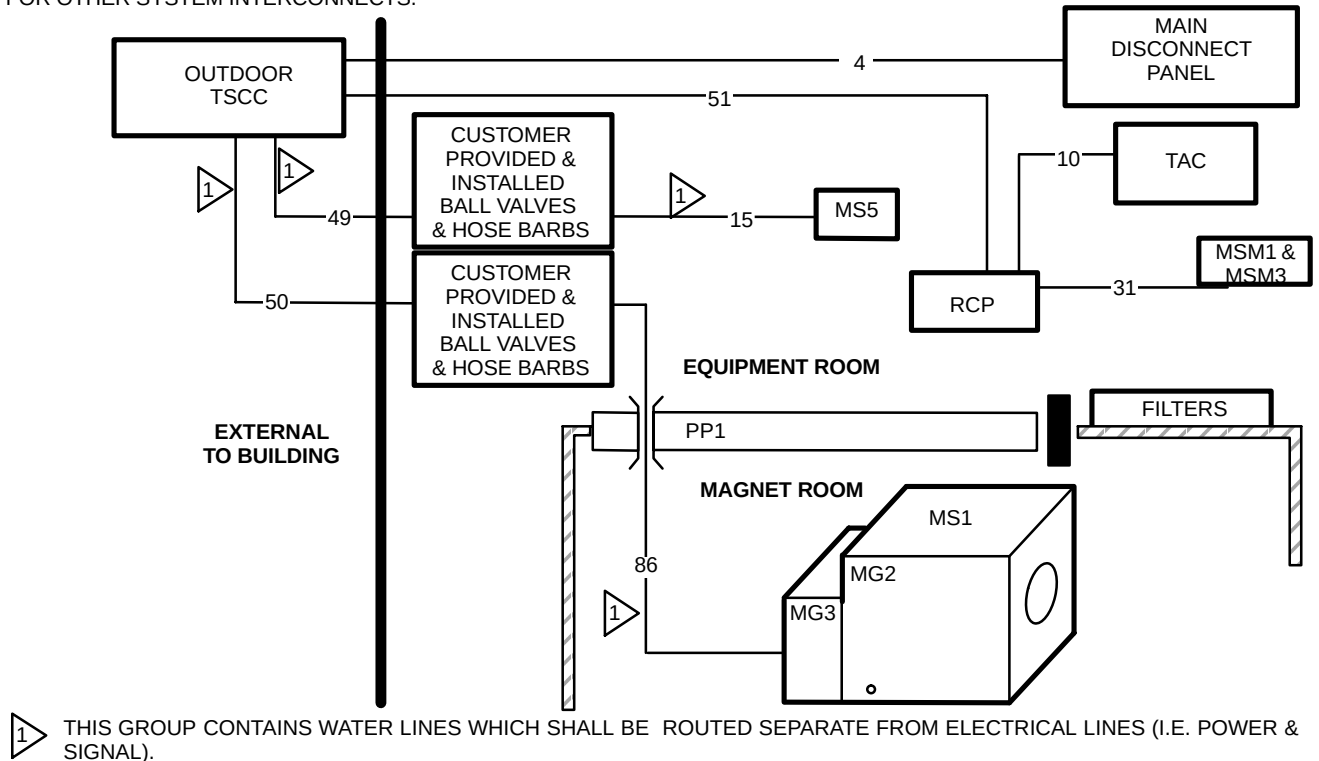
System Additional Interconnects

Illustration 6-3 shows the Outdoor TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-9 for specifics.

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



OUTDOOR TSCC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-3

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)

TABLE 6-9
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.²(mm²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Out- door TSCC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	RCP	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	Cus- tomer pro- vided ball valves & hose barbs for Shield/ Cryo Cooler	MS5	55 (16.7) minus takeup at hose barbs See Notes	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	Group 15 & 49 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements
31	0.10 (66)	MSM1	RCP	50 (15.24) minus takeup at both ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
49 SEE NOTE 1	—	Out- door TSCC	Cus- tomer ball valves & hose barbs for Shield/ Cryo Cooler	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	
● Customer to provide & install 0.75 (19 mm) ID copper lines for Shield/Cryo Cooler Compressor water cooling supply & return. ● Customer to provide & install in Equipment Room in end of each copper line a 0.75 in. (19 mm) ball valve & 0.5 in. (12.7 mm) hose barb. ● Groups 49 & 15 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements									
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).									
(Continued)									

6-2-13 M3088TA – Outdoor TSCC System Interconnects (Continued)

TABLE 6-9 (Continued)
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
50 SEE NOTE 1	—	Out-door TSCC	Cus- tomer ball valves & hose barbs for Gradi- ent Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	● Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. ● Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. ● Group 50 and 86 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements
51	1.097 (613)	Out-door TSCC	RCP	See NOTES	— — —	0.63 (16.0) 0.65 (16.5) 0.76 (15.9)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TSCC and height of connection at RCP. No cable takeup inside of TSCC or RCP.
86 SEE NOTE 1 & 2	1.666 (1078)	Cus- tomer pro- vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	● Group 86 and 50 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements ● These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.									

6-2-14 M3088TB – Indoor Water Cooled TSCC System Interconnects

Note

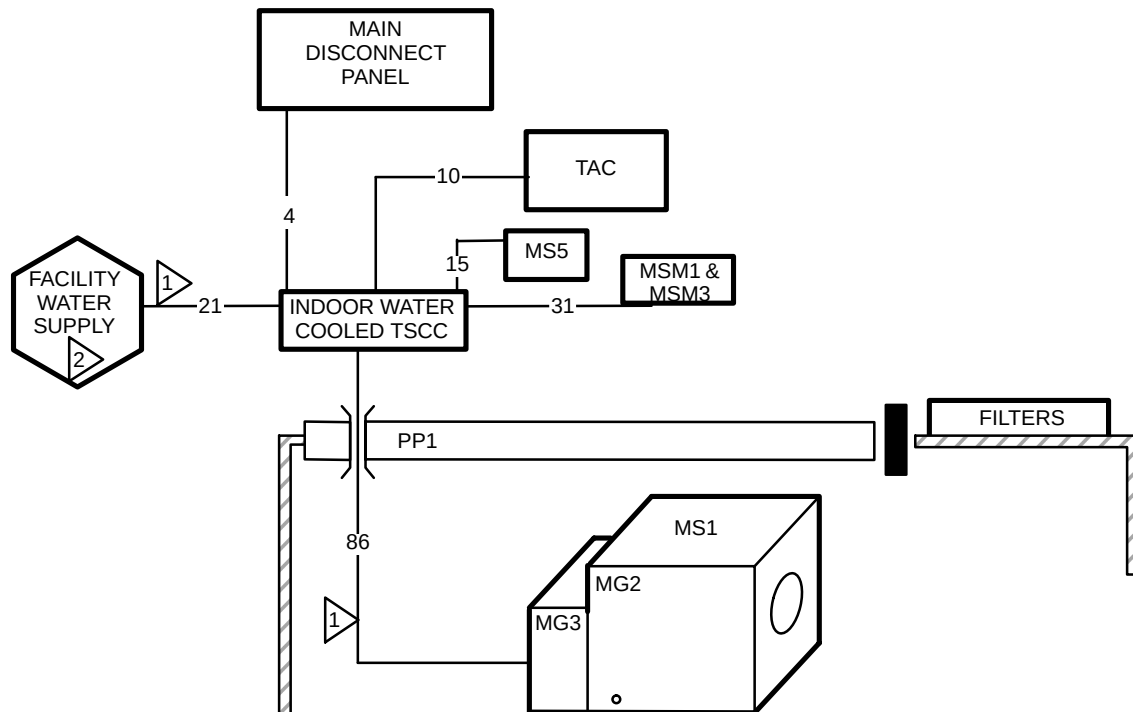
Refer to **Section 2-5-1 TSCC Configurations Siting Considerations** Table 2-4 listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 471 Pre installation Installation Operating LC 18MT/WC Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TSCC must be in accordance with local and national codes.

Illustration 6-4 shows the Indoor Water Cooled TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-10 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



- 1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- 2 FOR INDOOR WATER COOLED TSCC ONLY. REFER TO Section 4-4-3 **Water Cooled System Cooling Cabinet Configuration Requirements**

INDOOR WATER COOLED TSCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM

ILLUSTRATION 6-4

6-2-14 M3088TB – Indoor Water Cooled TSCC (Continued)

TABLE 6-10
INDOOR WATER COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	rubber hose rubber hose	rubber hose rubber hose	See Notes 3 & 4 for vertical separation limits & TSCC hoses/connections.
21 SEE NOTE 1	—	Facility Water Supply	TSCC	—	—	—	—	—	For Indoor Water Cooled TSCC ONLY: Customer furnished, refer to Section 4-4-3 Water Cooled System Cooling Cabinet Configuration Requirements & refer to Note 6
31	0.10 (66)	MSM1	TSCC	38 (11.6) minus takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1 & 2	1.666 (1078)	TSCC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 3 & 5 for vertical separation limits & TSCC hoses/connections.

- Note**
- 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).
 - 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.
 - 3 The Indoor Water Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.
 - 4 The Indoor Water Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2-13 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor.
 - 5 The Indoor Water Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2-13 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.
 - 6 Facility chilled water connections at TSCC are 1.5 inch (38.1 mm) FNPT. Customer to connect with 1.5 inch (38.1 mm) MNPT.

6-2-15 M3088TC – Indoor Air Cooled TSCC System Interconnects (Continued)

TABLE 6-11
INDOOR AIR COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	See Notes 3 & 4 for vertical separation limits & TSCC hoses/connections.
31	0.10 (66)	MSM1	TSCC	40 (12.2) minus takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1 & 2	1.666 (1078)	TSCC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 3 & 5 for vertical separation limits & TSCC hoses/connections.
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA. 3 The Indoor Air Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil. 4 The Indoor Air Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2-15 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor. 5 The Indoor Air Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2-15 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.									

6-2-16 M3088TE – Indoor Water Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2-5-2 TGWC Configurations Siting Considerations** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 469 Pre installation Installation Operating LTL – 10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TGWC must be in accordance with local and national codes.

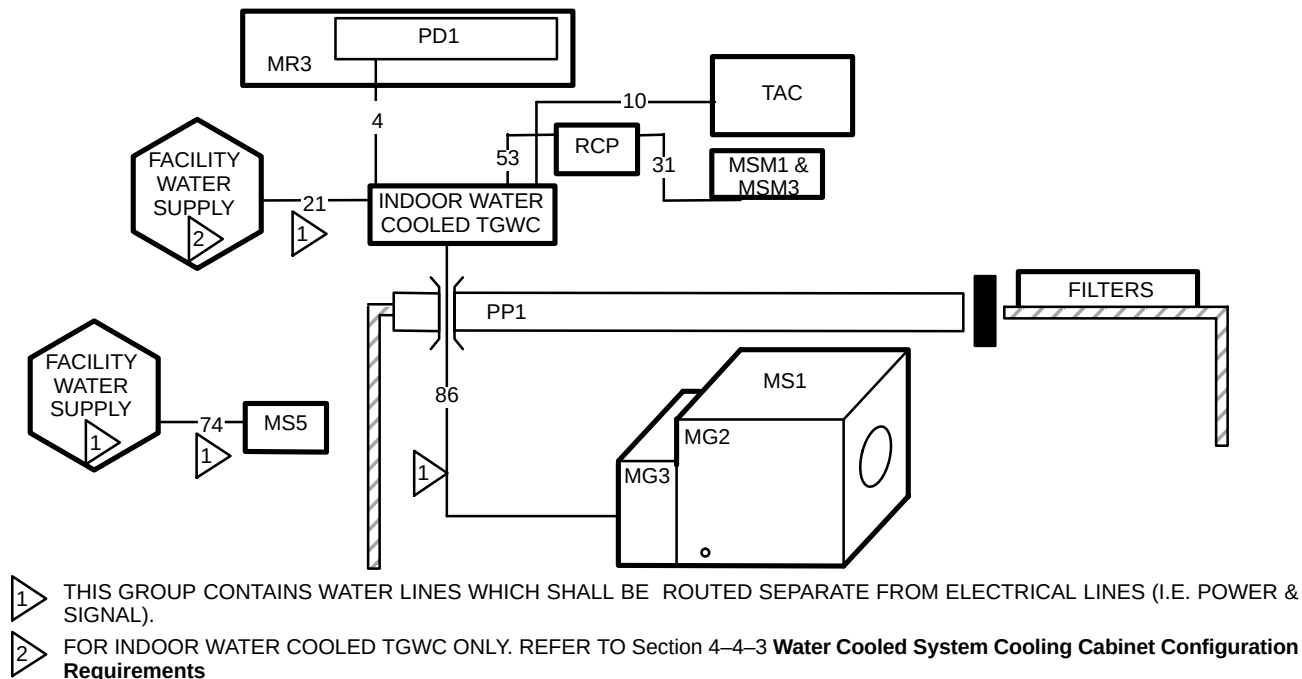
Illustration 6-6 shows the Indoor Water Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-12 for specifics.

Note

The Indoor Water Cooled TGWC and Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) of water lines. The TGWC vertical separation must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



INDOOR WATER COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-6

6-2-16 M3088TE – Indoor Water Cooled TGWC (Continued)

TABLE 6-12
INDOOR WATER COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	0.159 (103)	PD1 See Note 3	TGWC	23 (7.0)	—	0.45 (11.43)	Hard Wired	1.53x2.78 (38.9x70.6)	Ellis & Watts provided cable
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
21 SEE NOTE 1	—	Facility Water Supply	TGWC	—	—	—	—	—	For Indoor Water Cooled TGWC ONLY: Customer furnished, refer to Section 4-4-5 Water Cooled Twin Gradient Water Cooler (TGWC) Requirements
31	0.10 (66)	MSM1	RCP	50 (15.24) minus takeup at booth ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
53	0.157 (103)	Water Cooled TGWC	RCP	28 (8.53) minus takeup at RCP See Notes	—	0.258 (6.6) 0.258 (6.6) 0.258 (6.6)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	No cable takeup inside of RCP.
74	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to Sections 4-4-4 and 5).
86 SEE NOTE 1 & 2	1.666 (1078)	TGWC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 4 & 5 for vertical separation limits & TGWC hoses/connections.

- Note**
- 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).
 - 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.
 - 3 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).
 - 4 Indoor Water Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil.
 - 5 The Indoor Water Cooled TGWC has two 0.75 in. (19 mm) Snapfit OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2-20 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.

6-2-17 M3088TF – Outdoor TGWC System Interconnects**Note**

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2-5-2 TGWC Configurations Siting Considerations** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 474 Pre installation Installation Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for Outdoor TGWC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Single Loop Outdoor TGWC must be in accordance with local and national codes.**

Vertical Separation Requirements

- Dual Loop Outdoor TSCC must not exceed 100 ft (30.5 m) vertical separation from the Gradient Coil.

Gradient Coil Water Cooling Lines & Hoses Requirements

- Ellis & Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2-10 for location of fittings on TGWC. The Outdoor TGWC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
 - Total line length from the Outdoor TGWC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TGWC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
 - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduces 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
 - Copper lines must be insulated outdoors.

RCP Data Cables Requirements

- Ellis & Watts provides two data cables that connect between the RCP and the Outdoor TGWC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TGWC and RCP placement [total length minus height of connection at Outdoor TGWC and height of connection at RCP, no cable takeup inside of TGWC or RCP].

Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

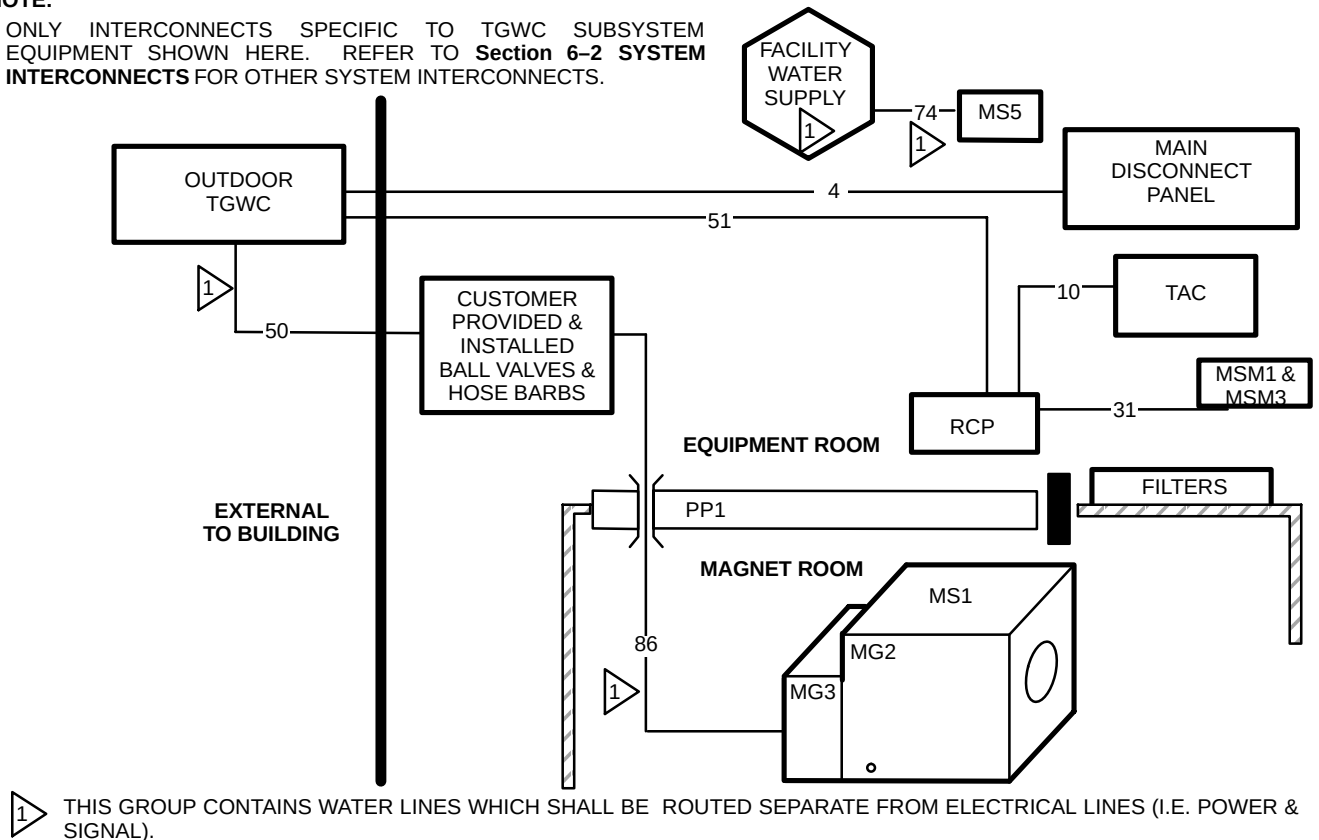
6-2-17 M3088TF – Outdoor TGWC System Interconnects (Continued)

System Additional Interconnects

Illustration 6-7 shows the Outdoor TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-13 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



OUTDOOR TGWC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM

ILLUSTRATION 6-7

6-2-17 M3088TF – Outdoor TGWC System Interconnects (Continued)

TABLE 6-13
OUTDOOR TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Out- door TGWC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.10 (66)	MSM1	RCP	60 (18.3) minus takeup at both ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
50 SEE NOTE 1	—	Out- door TGWC	Cus- tomer ball valves & hose barbs for Gradi- ent Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	● Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. ● Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. ● Group 50 and 86 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements
51	4.791 (3091)	Out- door TGWC	RCP	See NOTES	— —	1.59 (40.4) 1.89 (48.0)	1.97x2.25 (50.0x57.2) 1.75x2.25 (44.5x57.2)	1.97x2.25 (50.0x57.2) 1.75x2.25 (44.5x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TGWC and height of connection at RCP. No cable takeup inside of TGWC or RCP.
74 SEE NOTE 1	—	Water	MS5	—	—	—	—	—	Customer Furnished water lines (refer to Sections 4–4–4 and 5).
86 SEE NOTE 1 & 2	1.666 (1078)	Cus- tomer pro- vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	● Group 86 and 50 MUST NOT EXCEED limitations specified under Sec 6–2–13 sub–heading Gradient Coil Water Cooling Lines & Hoses Requirements ● These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA.									

6-2-18 M3088TG – Indoor Air Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

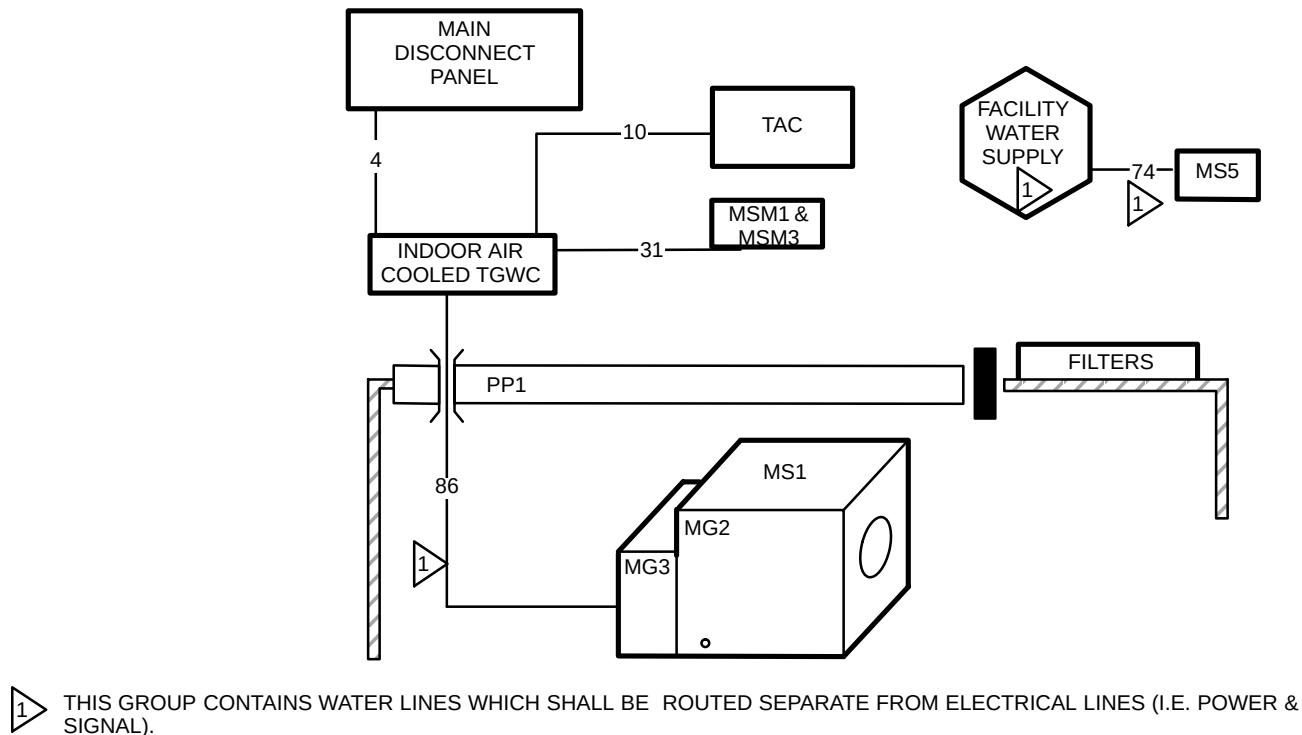
Refer to **Section 2-5-2 Fixed Site System TGWC Configurations Siting Considerations** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 473 Pre installation Installation Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Air Cooled TGWC must be in accordance with local and national codes.

Illustration 6-8 shows the Indoor Air Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6-14 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6-2 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



INDOOR AIR COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-8

6-2-18 M3088TG – Indoor Air Cooled TGWC (Continued)

TABLE 6-14
INDOOR AIR COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TGWC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.16 (103)	MSM1	TGWC	48 (14.6) minus takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
74	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to Sections 4-4-4 and 5).
86 SEE NOTE 1 & 2	1.666 (1078)	TGWC	MG2	80 (24.4)	— —	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 3 & 4 for vertical separation limits & TGWC hoses/connections.
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal). 2 GE provided water lines are part of Site Collector Kit: M3043TA, M3044TA, or M3063TA. 3 Indoor Air Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil. 4 The Indoor Air Cooled TGWC has two 0.75 in. (19 mm) Snpptite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2-22 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.									

6-2-19 M3088TH – GE Main Disconnect Panel

Table M3080TA system interconnects for the GE Main Disconnect Panel (MDP) are listed in Table 6-15. If customer provided MDP has been selected based on meeting **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts requisites then customer provided MDP MUST meet all MDP requirements, refer to **SECTION 5 – POWER REQUIREMENTS**.

TABLE 6-15
CABLES GE PROVIDED & CUSTOMER SUPPLIED FOR M3088TH

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	MDP	—	—	—	—	—	Customer Furnished. See Note 1.
2	MDP	PD1 See Note 2	—	—	—	—	—	Customer Furnished.
4	MDP/ PDU See Notes	TSCC/ TGWC See Notes	—	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine FROM connection.
Note 1 If low Voltage Step-Up Transformer Option (R4500AW or R4500BE) is used then customer furnished interconnects are required between facility power, transformer and MDP. 2 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).								

6-2-20 M3090AH – Horizon to TwinSpeed Upgrade System Electronics

M3090AH provides system interconnects for several assemblies mounted to the OW1 Table and routed internal to the OW1 Table cable tray.

6-2-21 M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet

Catalog M3090BB provides ACGD upgrade power and ground cables. Refer to Table 6-16 for cables details.

TABLE 6-16
CABLES PROVIDED BY CATALOG M3090BB & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-2-22 M3090EB – EXCITE UPGRADE KIT

M3090EB provides standard length system interconnects for fixed site listed in Table 6-17.

TABLE 6-17
CABLES PROVIDED BY CATALOG M3090BB & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	1034	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 or 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1025	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1026	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1027	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1028	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1029	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
				1030	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
				1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
45	PP1	MG2/3	30 (9.15)	1037	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1038	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1039	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1040	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1041	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1042	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-2-22 M3090EB – EXCITE UPGRADE KIT (Continued)

TABLE 6-17 (Continued)
CABLES PROVIDED BY CATALOG M3090BB & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5)	1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
				1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	

6-2-23 M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet

Cables provided by catalogs M3090BC are the same as catalogs M3090BB. Refer to Table 6-16 for cables details.

6-2-24 M3090TA – Twin Upgrade For LCC Horizon Fixed & T/R

Catalog M3090TA provides fixed site and Transportable/Relocatable system interconnect cables for Horizon Magnet Enclosure configuration to Signa 1.5T TwinSpeed. Refer to Table 6-18 for cables details.

TABLE 6-18
CABLES PROVIDED BY CATALOG M3090TA & M3090TE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	MR3	TAC	SEE ROW NOTES	976	1.21 (30.7)	Hard Wired	Hard Wired	Runs 976, 977, & 978 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
				977	1.21 (30.7)	Hard Wired	Hard Wired	
				978	1.21 (30.7)	Hard Wired	Hard Wired	
9	PD1 SEE NOTE1	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	
10	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
11	TAC	PP1	41 (12.5)	991	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
12	TAC	PP1	SEE ROW NOTES	979	1.21 (30.7)	Hard Wired	Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 75 ft (22.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
				980	1.21 (30.7)	Hard Wired	Ring Terminal	
				981	1.21 (30.7)	Hard Wired	Ring Terminal	
				988	0.26 (6.6)	Ring Terminal	Ring Terminal	Ground wire, Ground wire, 65 ft (19.8 m) actual length.
13	TAC	MG3	80 (24.4)	—	1.27 (32.3)	Flexible Line	Flexible Line	Run is routed through waveguide in PP1, cut to length at site.
31	MSM1	RCP/ TSCC	SEE ROW NOTES	941	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6–2–13 through 6–2–18) for details & to determine TO connection.
38	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
45	PP1	MG3	30 (9.15)	300	0.44 (11.18)	2.30x2.00 (57.2x50.8)	Ring Terminals	
				990	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).								
(Continued)								

6-2-24 M3090TA – Twin Upgrade For LCC Horizon Fixed & T/R (Continued)

TABLE 6-18 (Continued)
CABLES PROVIDED BY CATALOG M3090TA & M3090TE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
85	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
				983	1.21 (30.7)	Ring Terminals	Hard Wired	
				984	1.21 (30.7)	Ring Terminals	Hard Wired	
				985	1.21 (30.7)	Ring Terminals	Hard Wired	
				986	1.21 (30.7)	Ring Terminals	Hard Wired	
				987	1.21 (30.7)	Ring Terminals	Hard Wired	
86 SEE NOTE 2	TSCC/ TGWC	MG2	SEE ROW NOTES	—	—	—	—	Refer to appropriate section for TSCC or TGWC interconnects (Section 6-2-13 through 6-2-18) for details & to determine TO connection.
Note 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).								

6-2-25 M3090TE – Twin Upgrade Covers From WideOpen For Fixed & T/R Sites

Cables provided by catalogs M3090TE are the same as catalogs M3090TA. Refer to Table 6-18 for details.

6-3 ADVANTAGE WORKSTATION INTERCONNECTS

Refer to Direction 2261309-100 Advantage Workstation Ultra60 Pre-Installation for AW interconnect information.

6-4 INTERCONNECTS RE-ROUTED BY ACGD UPGRADES

ACGD Gradients upgrades replace the existing system PDU (Full-size, Compact, or PDU in RF/PDU Cabinet) with the PDU located in the lower portion of the ACGD/PDU Cabinet. Therefore, all connections to the PDU may need to be re-routed depending on site layout of the upgraded system.

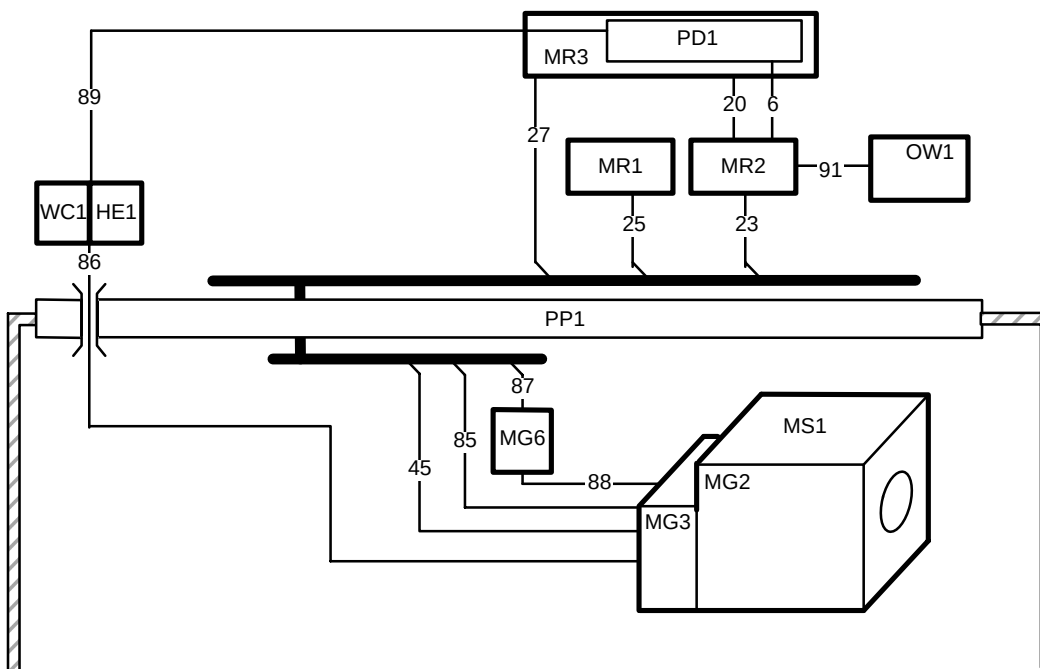
Note

GE Installation Specialist & site Field Engineer need to consider impact of upgrade site layout on re-routed cables especially if any cables (e.g. power cables) were cut to length for current installed system.

6-5 INTERCONNECTS REMOVED BY SIGNA 1.5T TWINSPEED WITH EXCITE TECHNOLOGY UPGRADES

6-5-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR System With LCC Magnet & ACGD & SRF Cabinets

Illustration 6-9 and Table 6-19 lists the existing Signa MR/i, CV/i, NV/i & Signa LX, CVMR system with ACGD/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed With EXCITE Technology configuration.



SIGNA MR/i, CV/i, NV/i OR SIGNA LX WITH ACGD/PDU CABINET TO TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE
REMOVED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-9

**6-5-1 Signa MR*li*, CV*li*, NV*li* Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-19
SIGNA MR*li*, CV*li*, NV*li* OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
6	MR3/ PD1 See Note 1	MR2	22 (6.70)	972	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40) See Note 2	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
(Continued)								

**6-5-1 Signa MRIi, CVIi, NVIi Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-19 (Continued)
SIGNA MRIi, CVIi, NVIi OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
(Continued)								

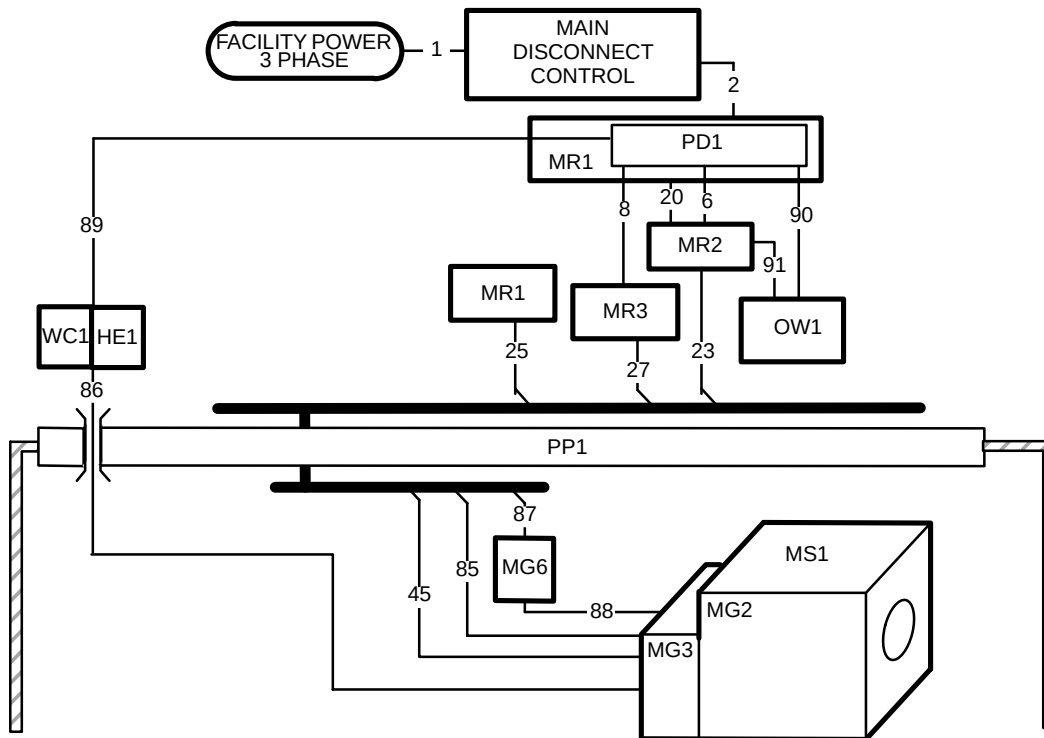
**6-5-1 Signa MR*i*, CV*i*, NV*i* Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-19 (Continued)
SIGNA MR*i*, CV*i*, NV*i* OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR <i>i</i> , CV <i>i</i> , & NV <i>i</i>	MR3/ PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	MR3/ PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
91	MR2	OW1	51 (15.5)	836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4.7 in. (120 mm).
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1). 2 M3044TA may have been ordered to increase usable length of Group 25 to 51 ft (15.5 m) except for Runs 235 & 236 which may have been Special Order Inquiry (SOI) to provide longer length.								

6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinets

Illustration 6-10 and Table 6-20 lists the existing Signa MR/i, CV/i, NV/i or LX system with SGD & RF/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed With EXCITE Technology configuration.



**SIGNA HORIZON LX WITH SGD & RF/PDU CABINET TO TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE
REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-10**

**TABLE 6-20
SIGNA MR/i, CV/i, NV/i OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE**

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR1/ PD1	MR2	22 (6.70)	030 972	0.92 (23.4) 0.36 (9.14)	Ring Terminals 1.30x2.00 (33.5x50.8)	Ring Terminals 1.30x2.00 (33.5x50.8)	Power cable, #8 AWG 5 wires
(Continued)								

**6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-20 (Continued)
SIGNA MR/II, CV/II, NV/II OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	

**6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-20 (Continued)
SIGNA MR/II, CV/II, NV/II OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	

(Continued)

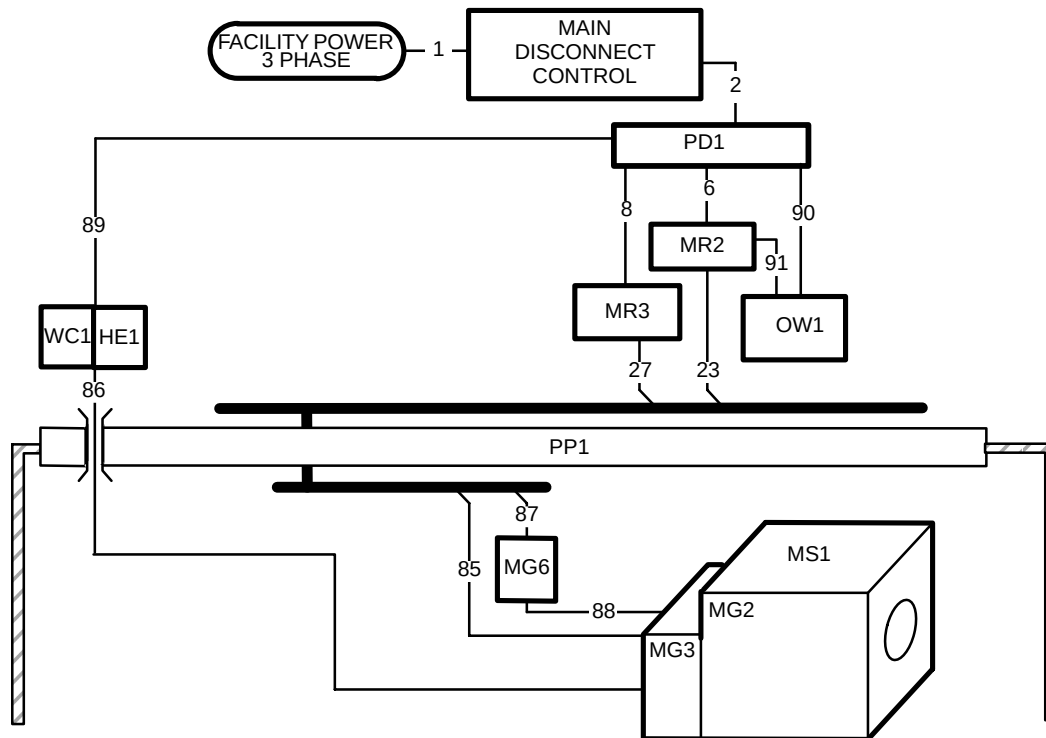
**6-5-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-20 (Continued)
SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY
TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).

6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets

Illustration 6-11 and Table 6-21 lists the existing Signa Horizon LX system with SGD & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed With EXCITE Technology configuration.



**SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET TO TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE
REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-11**

**TABLE 6-21
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE**

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR1/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
				972	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
(Continued)								

**6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-21 (Continued)
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	

**6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-21 (Continued)
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
(Continued)								

**6-5-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-21 (Continued)
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED
BY TWINSPEED WITH EXCITE TECHNOLOGY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	836	0.33 (8.3)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).

6-5-4 Signa LX & CVMR Without LCC Magnet Installed**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed With EXCITE Technology upgrade for Signa LX or CVMR (with ACGD or SGD) systems which currently do NOT have an LCC Magnet both the magnet and system electronics upgrades, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-5-5 Signa Horizon 5.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed With EXCITE Technology upgrade for Signa Horizon will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

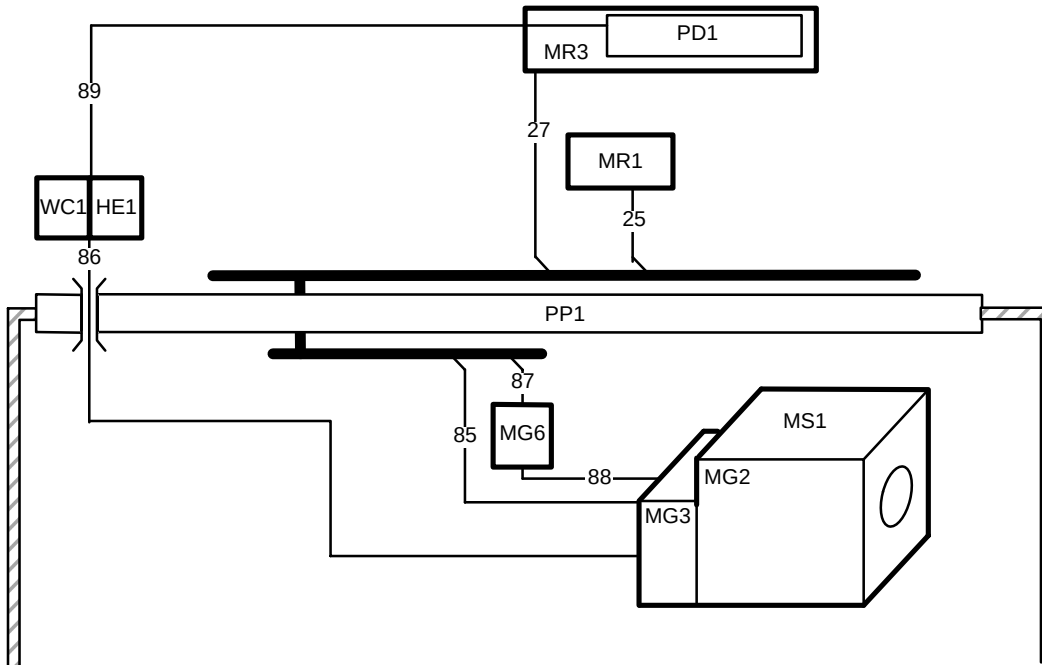
6-5-6 Signa Advantage 5.X & 4.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed With EXCITE Technology upgrade for Signa Advantage 5.X and 4.X will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-6 INTERCONNECTS REMOVED BY SIGNA 1.5T TWINSPEED UPGRADES**6-6-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR System With LCC Magnet & ACGD & SRF Cabinets**

Illustration 6-12 and Table 6-22 lists the existing Signa MR/i, CV/i, NV/i & Signa LX, CVMR system with ACGD/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed configuration.



SIGNA MR/i, CV/i, NV/i OR SIGNA LX WITH ACGD/PDU CABINET TO TWINSPEED UPGRADE REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-12

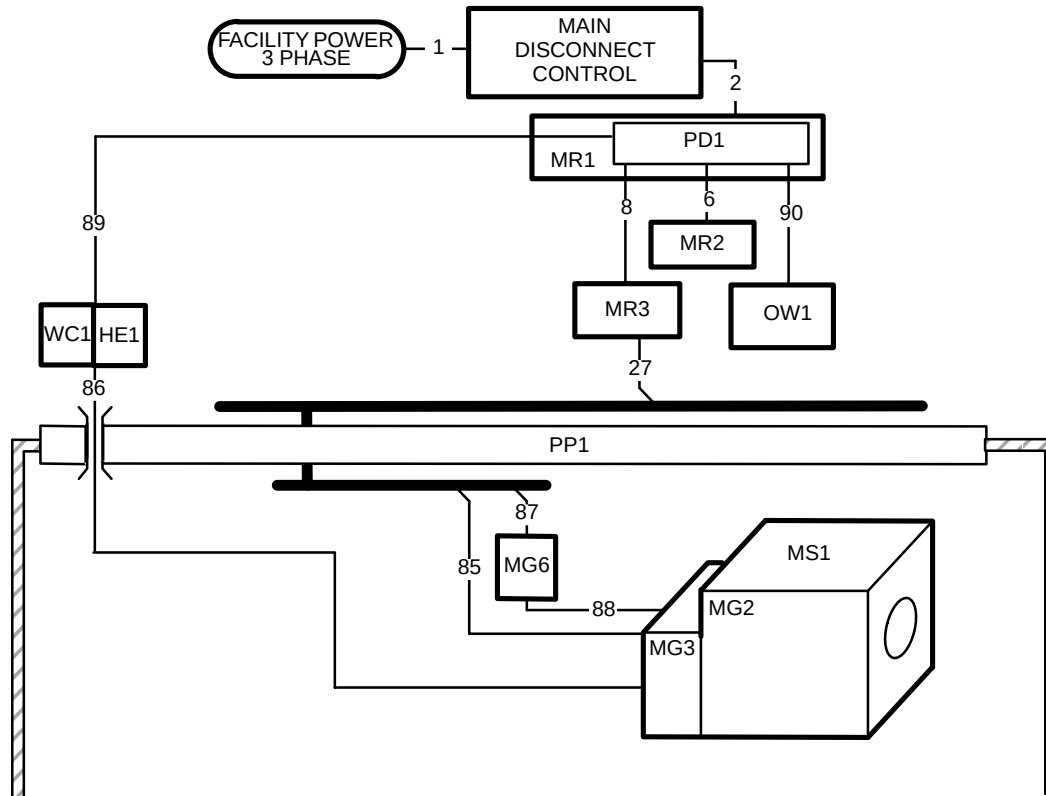
**6-6-1 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & ACGD & SRF Cabinets
(Continued)**

TABLE 6-22
SIGNA MR//, CV//, NV// OR SIGNA LX WITH ACGD/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water

6-6-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinets

Illustration 6-13 and Table 6-23 lists the existing Signa MR/i, CV/i, NV/i or LX system with SGD & RF/PDU Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed configuration.



SIGNA HORIZON LX WITH SGD & RF/PDU CABINET TO TWINSPEED UPGRADE REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-13

TABLE 6-23

SIGNA MR/i, CV/i, NV/i OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR1/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
(Continued)								

**6-6-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

TABLE 6-23 (Continued)

SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

(Continued)

**6-6-2 Signa MR/i, CV/i, NV/i Or Signa LX System With LCC Magnet & SGD & RF/PDU Cabinet
(Continued)**

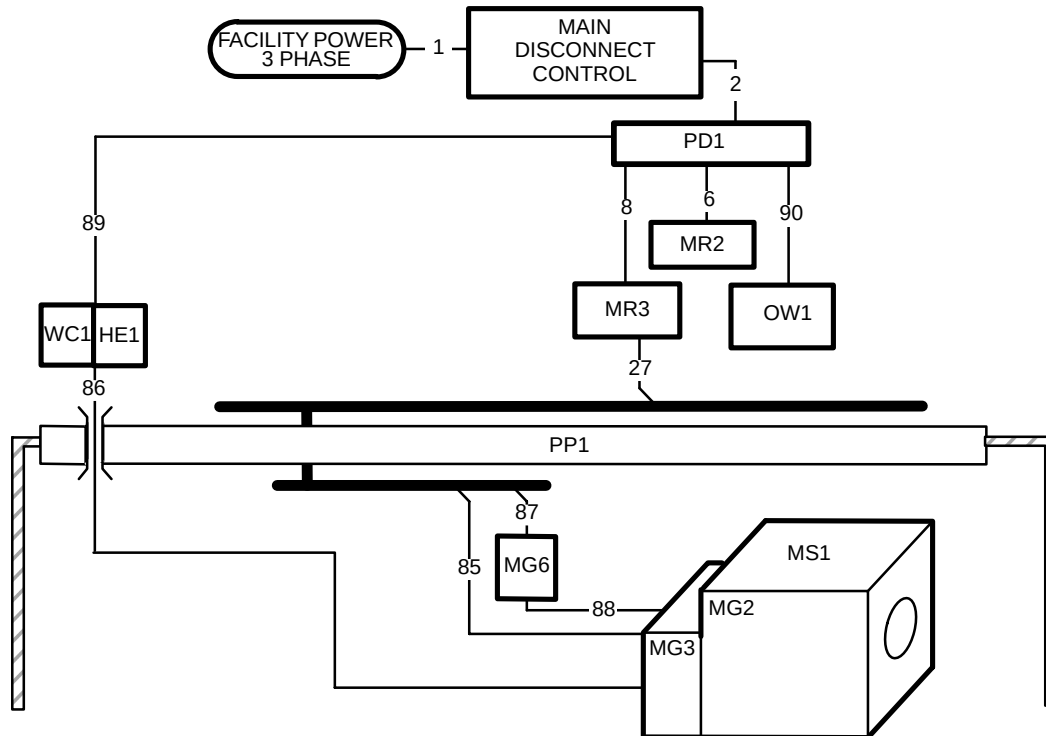
TABLE 6-23 (Continued)

SIGNA MR//, CV//, NV// OR SIGNA WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets

Illustration 6-14 and Table 6-24 lists the existing Signa Horizon LX system with SGD & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa 1.5T TwinSpeed configuration.



SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET TO TWINSPEED UPGRADE REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-14

TABLE 6-24

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY TWINSPEED UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
(Continued)								

**6-6-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-24 (Continued)

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
86	WC1 or HE1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See row Notes	786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
88	MG6	MG3	50 (15.2) minus takeup, See row Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses were cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-7, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

(Continued)

**6-6-3 Signa MR/i, CV/i, NV/i Or Signa LX With LCC Magnet & SGD & RF/Penetration Cabinets
(Continued)**

TABLE 6-24 (Continued)

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
89 MR/i, CV/i, & NV/i	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
89 LX & CVMR	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-4 Signa LX & CVMR Without LCC Magnet Installed**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed upgrade for Signa LX or CVMR (with ACGD or SGD) systems which currently do NOT have an LCC Magnet both the magnet and system electronics upgrades, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-6-5 Signa Horizon 5.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed upgrade for Signa Horizon will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-6-6 Signa Advantage 5.X & 4.X System**Note**

Signa 1.5T TwinSpeed upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the TwinSpeed Upgrade.

The Signa 1.5T TwinSpeed upgrade for Signa Advantage 5.X and 4.X will upgrade to an 1.5T LCC Magnet and the TwinSpeed system electronics, the site collectors catalogs and the magnet catalogs, listed in prior Section 6 – SYSTEM INTERCONNECTS subsections, provide all new GE system interconnects.

6-7 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2-9, COMPONENT DIMENSIONS. Signa 1.5T TwinSpeed several changes to the cabling and hoses routed through the Rear Pedestal cable access areas of the Magnet Enclosure. These changes include the

- elimination of 1 Blower Box hose (6.5 inch (165 mm) OD hose)
- installation of 3 additional Gradient cables
- addition of a vacuum hose and a vacuum sensor cable for the Quiet Technology Enclosure
- 2 existing water hoses for Body Coil are replaced with 2 new larger water hoses.

The remainder of the Magnet Enclosure system interconnect cables are also still routed through the Rear Pedestal cable access. The existing cable access area utilized to site cable routing should be sufficient for the upgrade system cabling. Illustration 6-15 shows one example of Blower Box set-up with flexible hoses route separate from other interconnects.

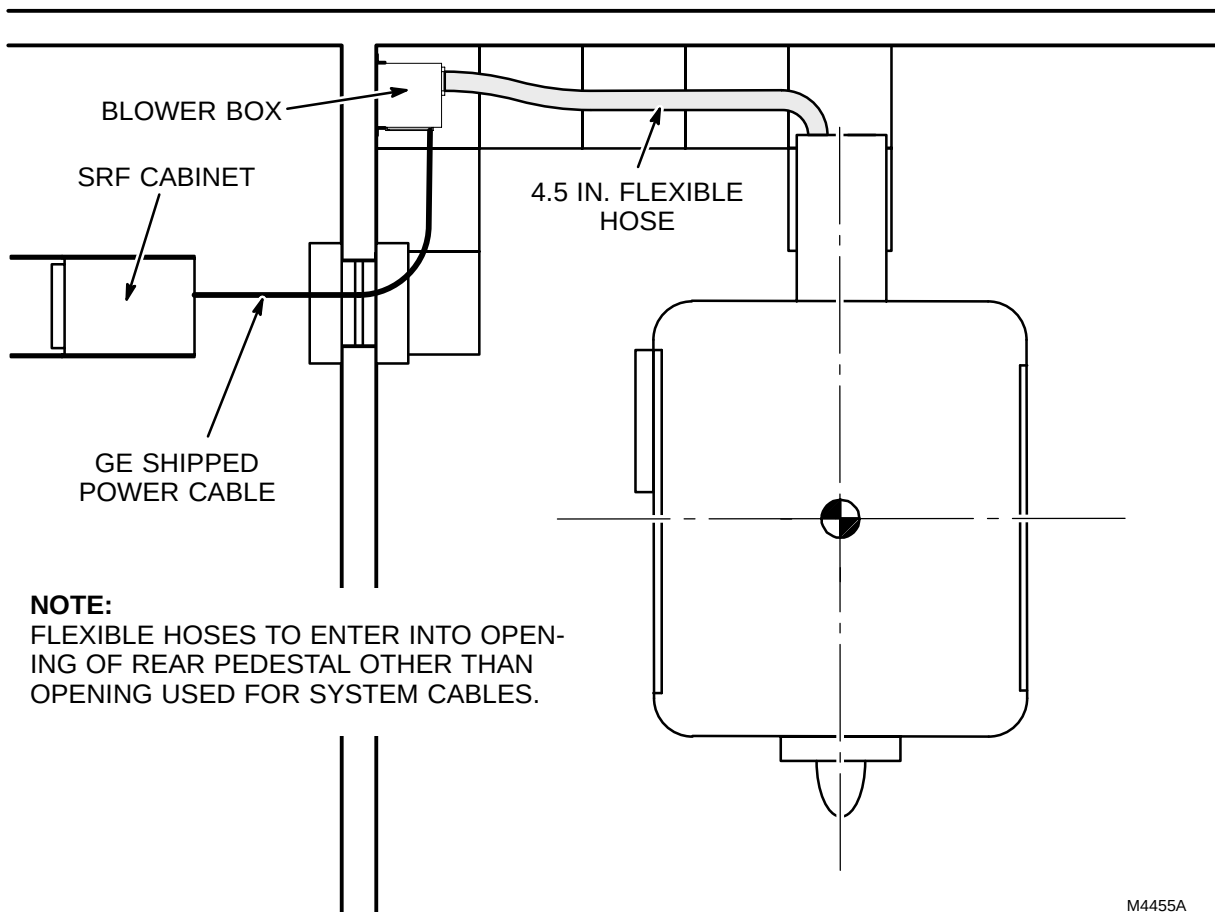
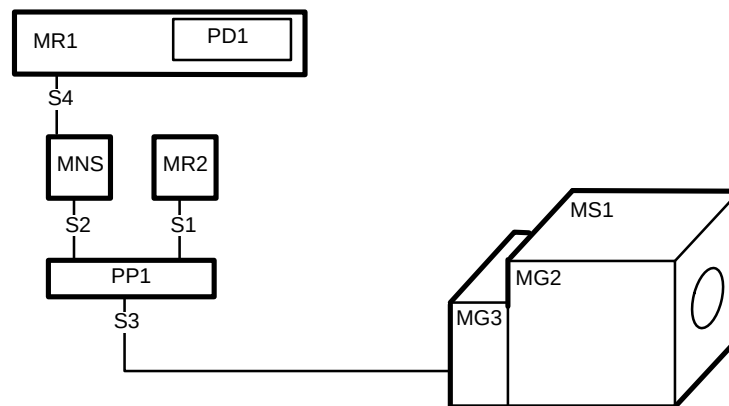
**EXAMPLE OF BLOWER BOX SET-UP (PLAN VIEW)**

ILLUSTRATION 6-15

6-8 SPECTROSCOPY OPTION INTERCONNECTS

6-8-1 M1033MD – MNS Kit For Signa LX Systems

Illustration 6-16 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-25 contains the cable data for these interconnects.



CATALOG M1033MD SPECTROSCOPY OPTION SYSTEM INTERCONNECTS

ILLUSTRATION 6-16

6-8-1 M1033MD – MNS Kit For Signa LX Systems (Continued)

TABLE 6-25
CABLES PROVIDED BY CATALOG M1033MD

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S2	0.41 (267)	PP1	MNS	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6-3 Group 25.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
					473	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6-8-2 M3090DB – 5.5 MNS Upgrade Kit For ACGD Site

Illustration 6-17 shows the system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-26 contains the cable data for these interconnects.

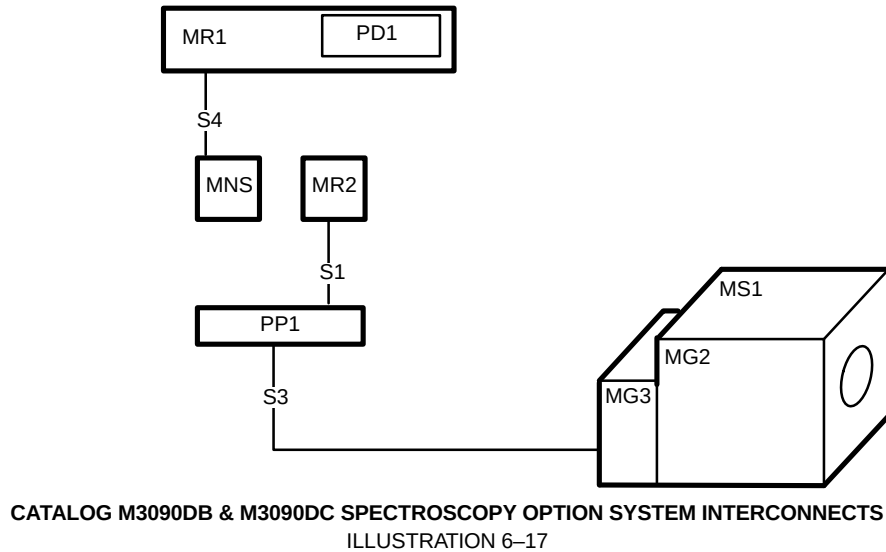


TABLE 6-26
CABLES PROVIDED BY CATALOG M3090DB & M3090DC

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6-8-3 M3090DC – LX MNS Upgrade Kit For ACGD Site

Cables provided by catalog M3090DC are the same as catalog M3090DB. Refer to Illustration 6-17 and Table 6-26 cables details.